

Abstract Linear Algebra

Prof. Lanier

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Pick up a handout. Try those warm-up problems.

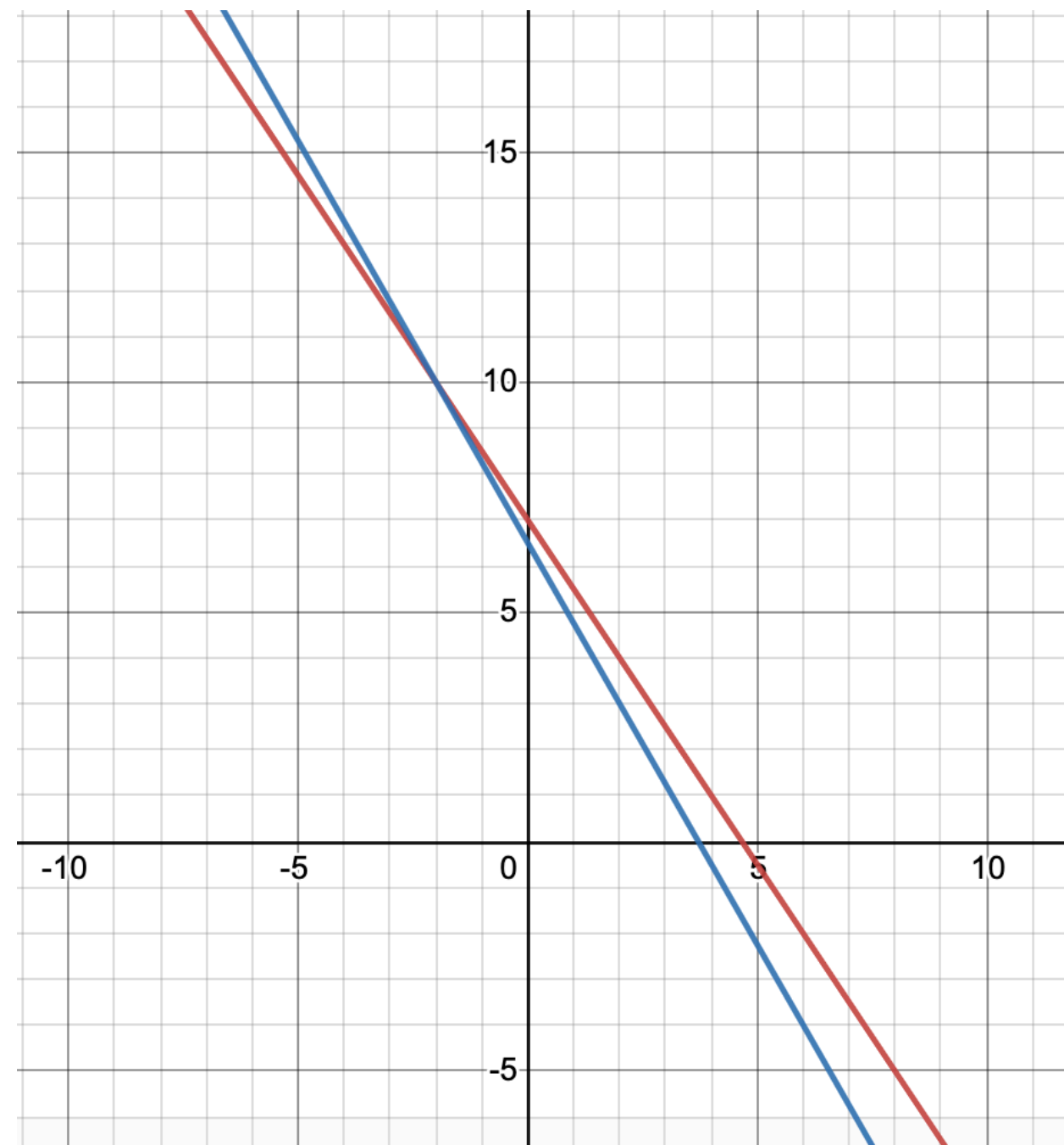
Next up:

- This week: Chapter 3 of *LADW*; Today: wrapping up Ch 2 (except 2.8).
- Quizzes 4 & 5 on Friday (finding inverses, solving systems to vector form)
- Homework 3 is due Friday / Saturday.
- HW2 debrief today 3:30-4:30pm in Eckhart 207A
- First problem swap problems due on Wednesday.

system of linear equations

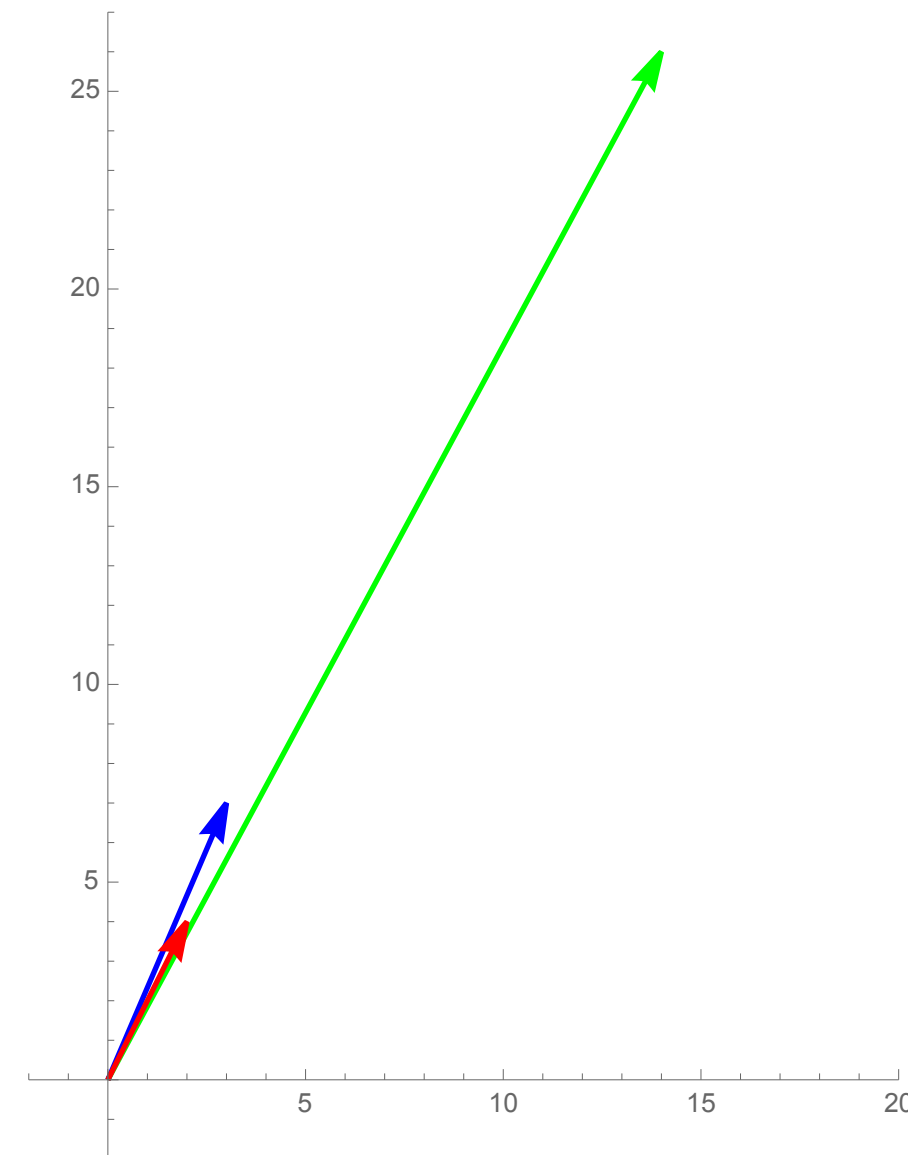
$$3x + 2y = 14$$

$$7x + 4y = 26$$



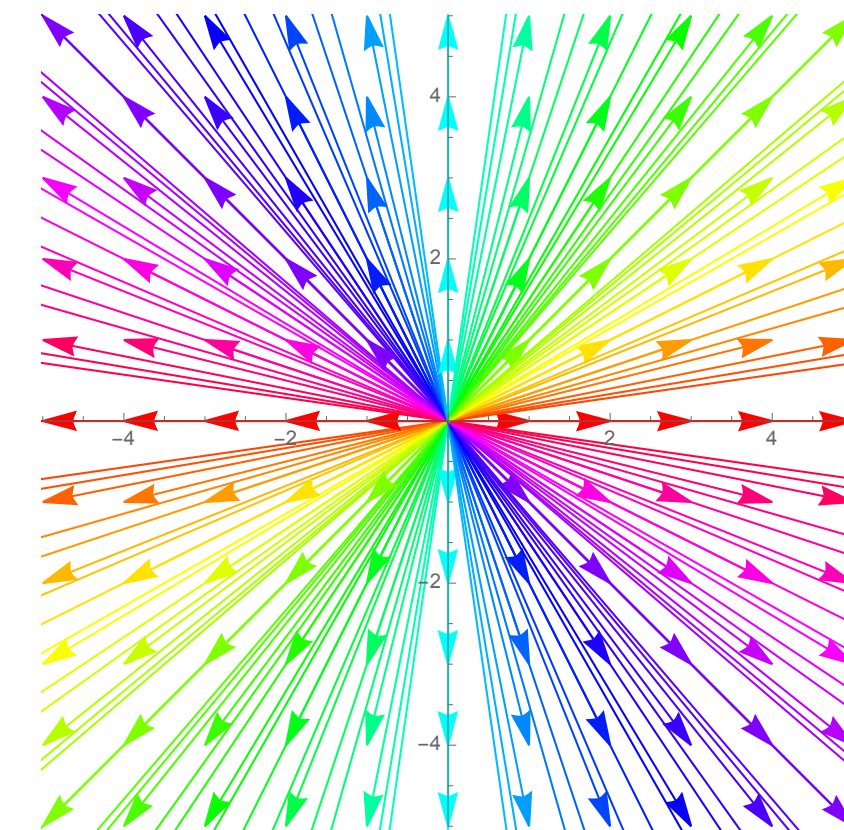
vector equation

$$x \begin{pmatrix} 3 \\ 7 \end{pmatrix} + y \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 14 \\ 26 \end{pmatrix}$$

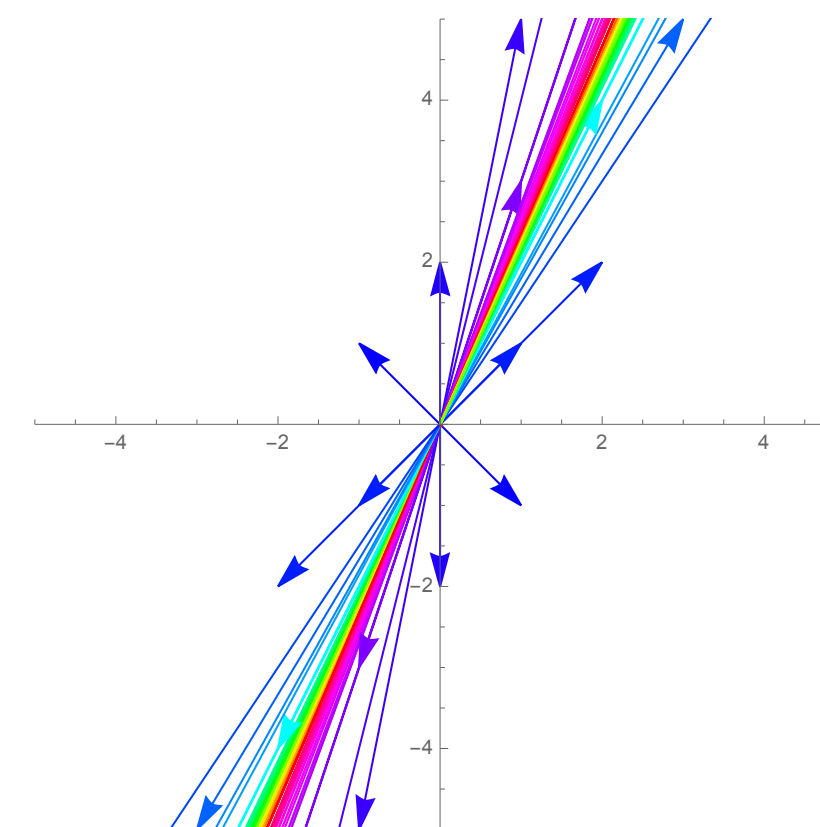


matrix equation

$$\begin{pmatrix} 3 & 2 \\ 7 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 14 \\ 26 \end{pmatrix}$$



$$\begin{pmatrix} 3 & 2 \\ 7 & 4 \end{pmatrix}$$

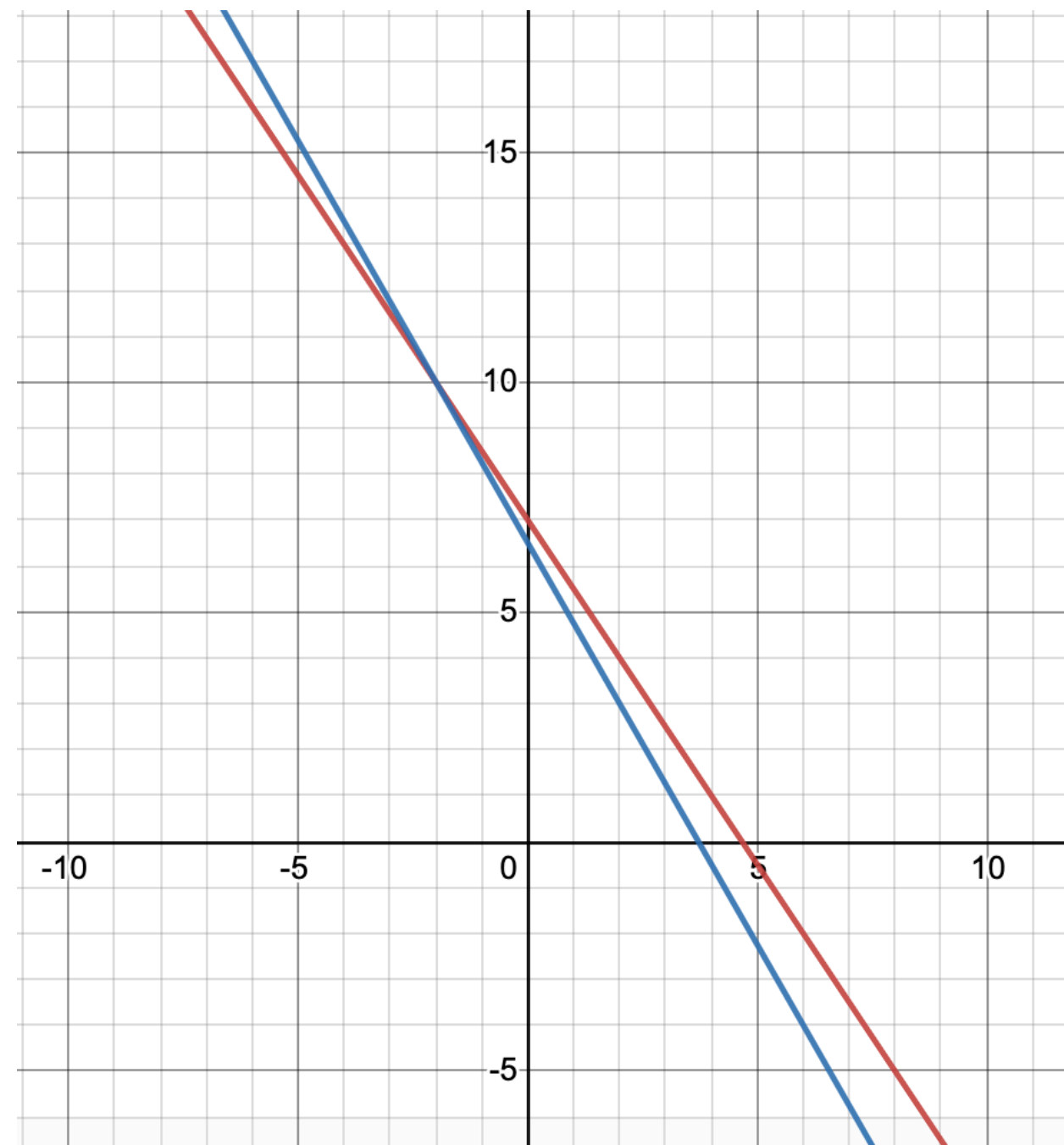


When does a solution exist?
When is the solution unique?

system of linear equations

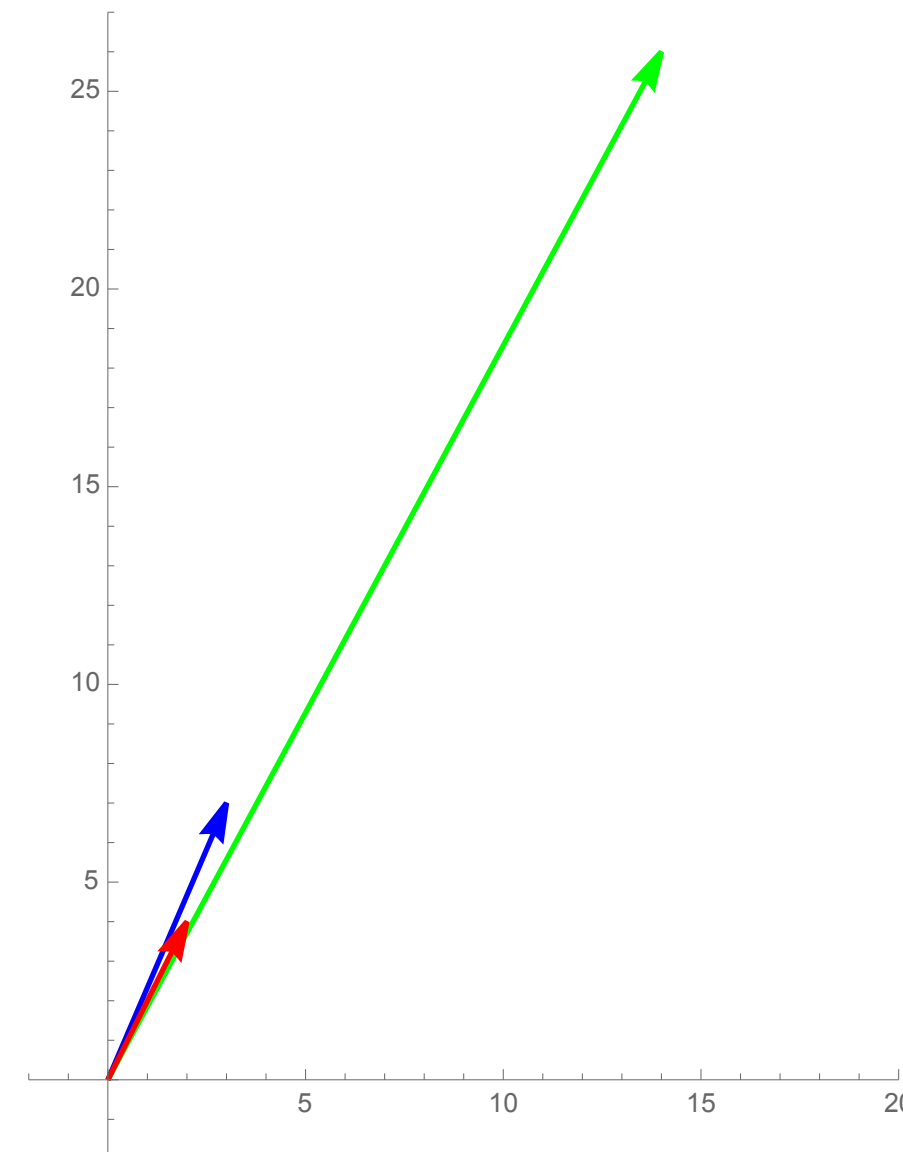
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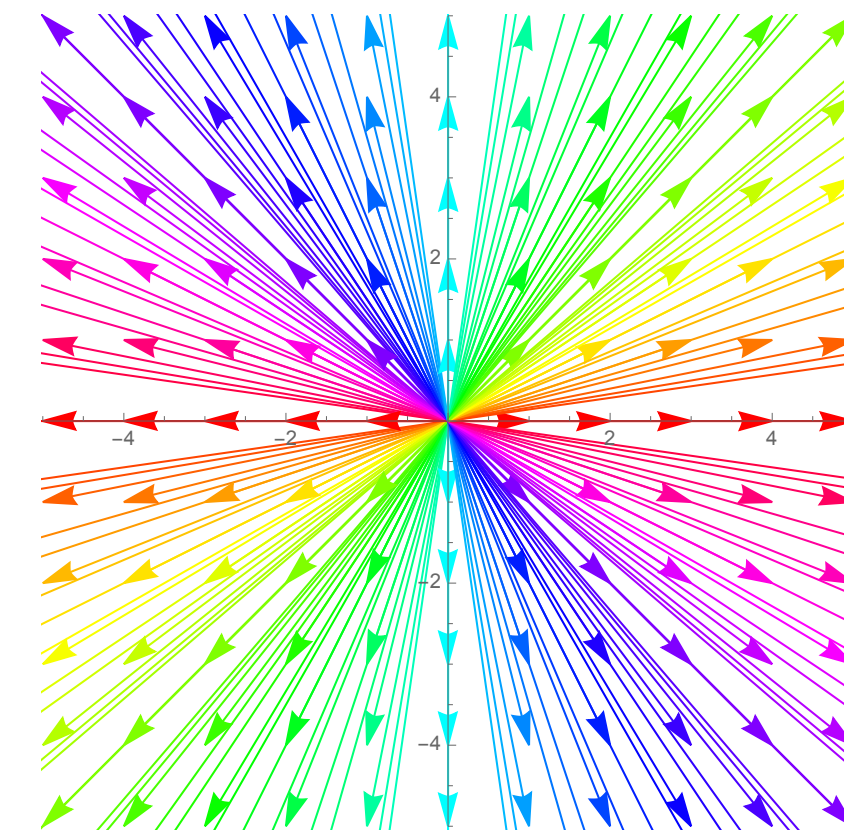
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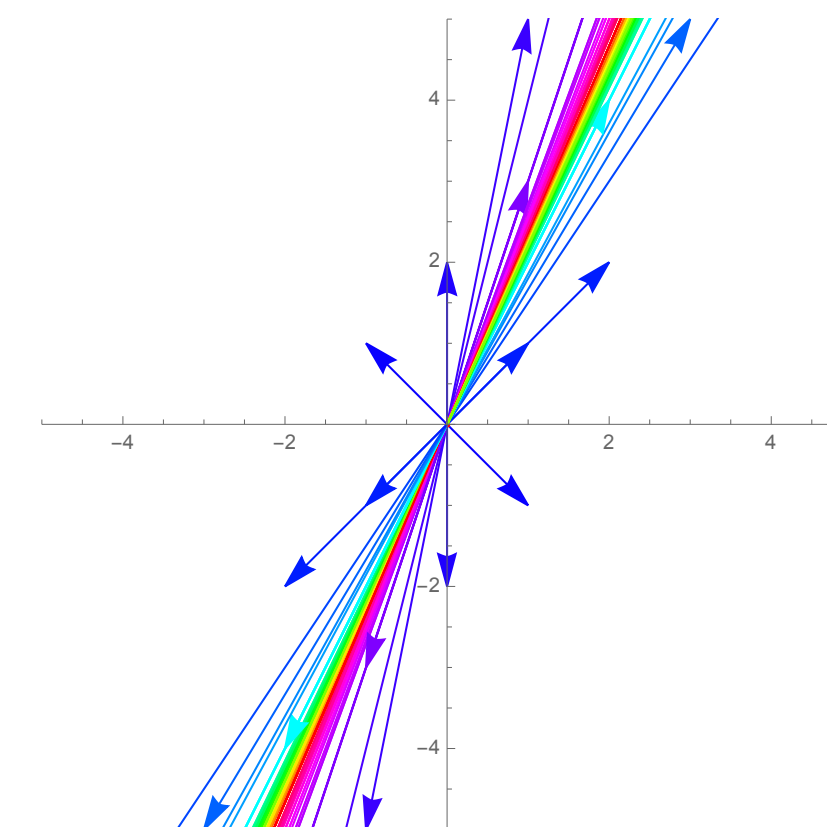


matrix equation

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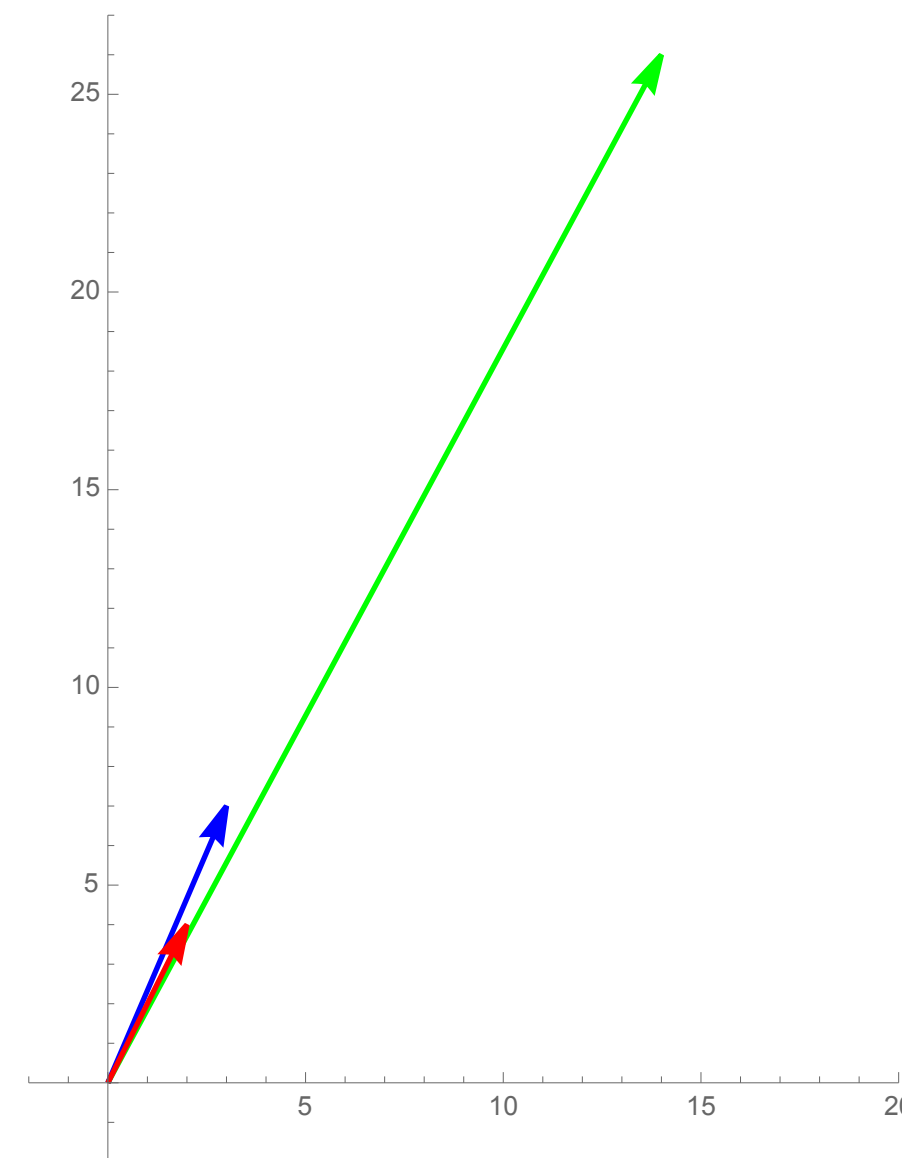


When does a solution exist?
When is the solution unique?

We can decide
by row reduction.

vector equation

$$x \begin{pmatrix} 3 \\ 7 \end{pmatrix} + y \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \boxed{?}$$

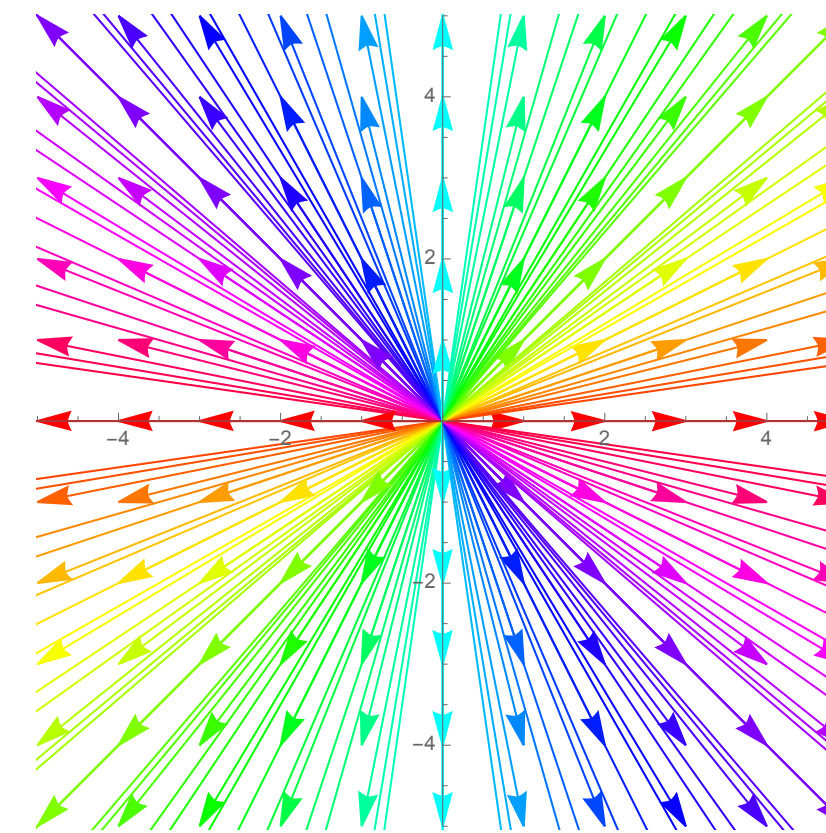


the vectors span
the vector space

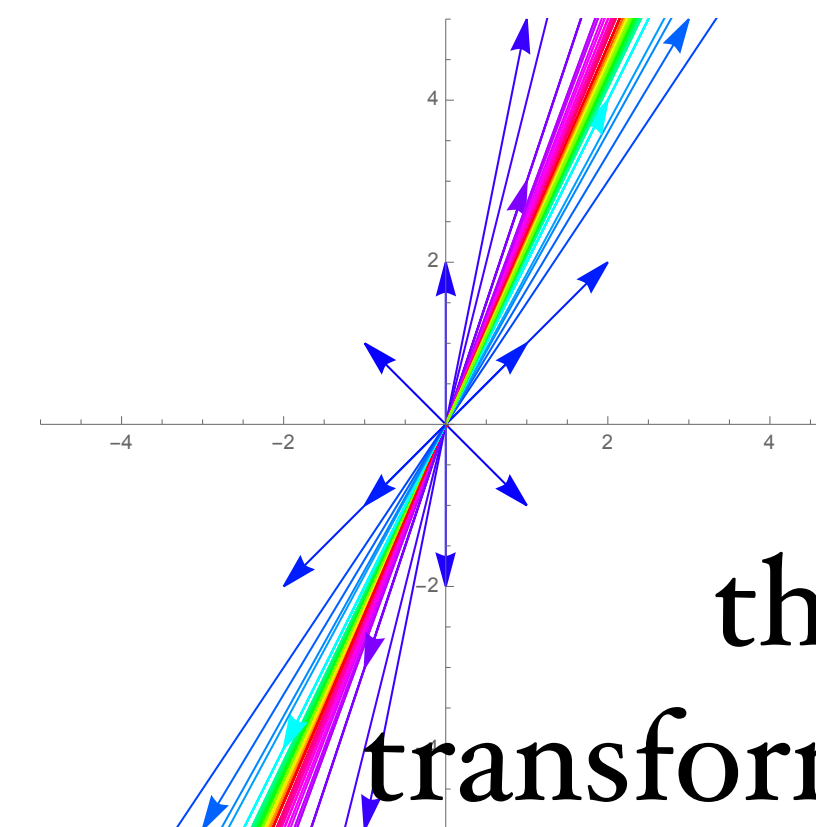
Solution *always* exists if:

matrix equation

$$\begin{pmatrix} 3 & 2 \\ 7 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \boxed{?}$$



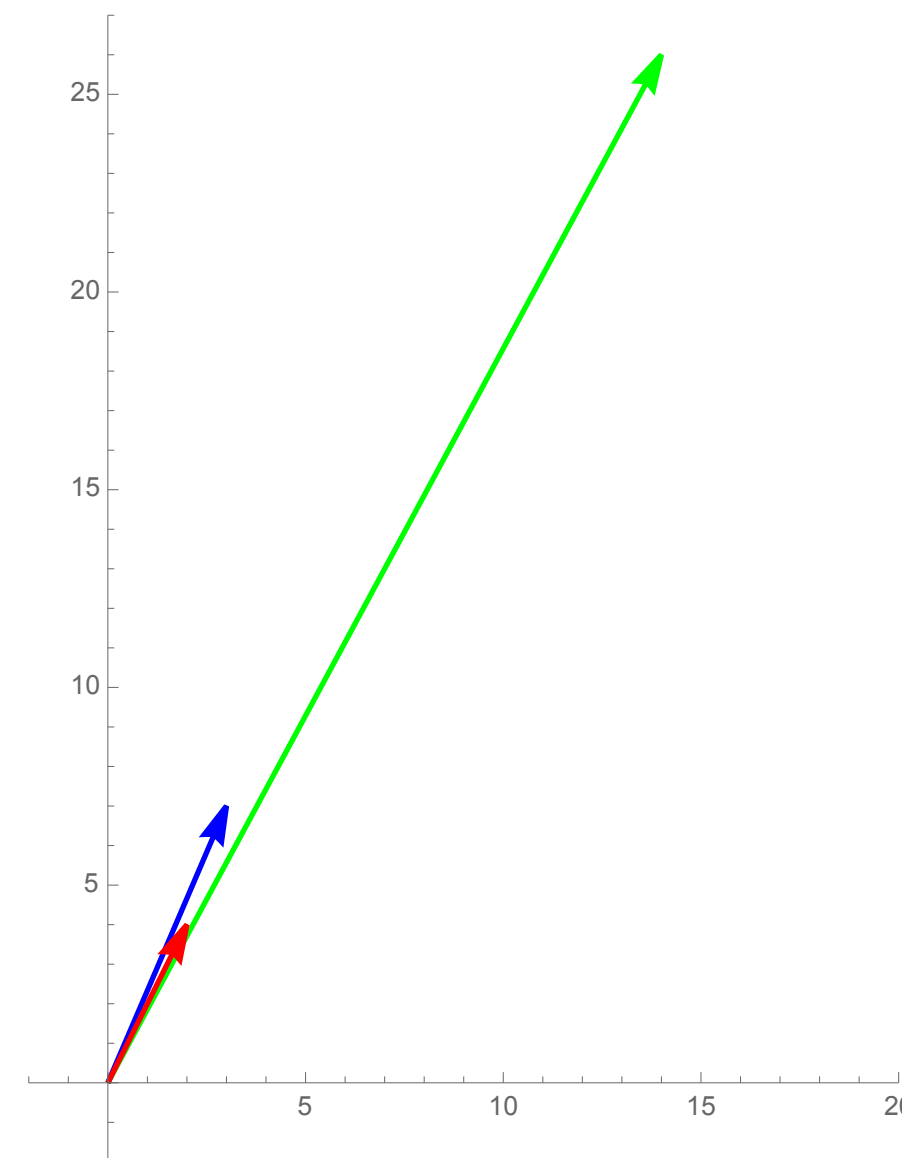
$\downarrow \begin{pmatrix} 3 & 2 \\ 7 & 4 \end{pmatrix}$



the linear
transformation is onto

vector equation

$$x \begin{pmatrix} 3 \\ 7 \end{pmatrix} + y \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \text{?}$$

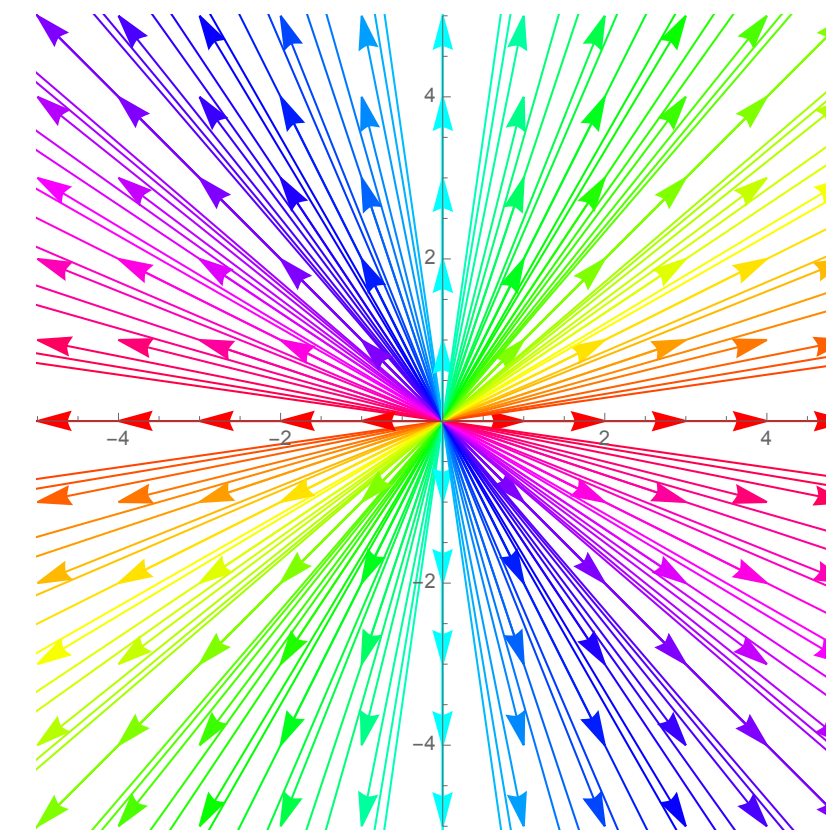


the vectors are
linearly independent

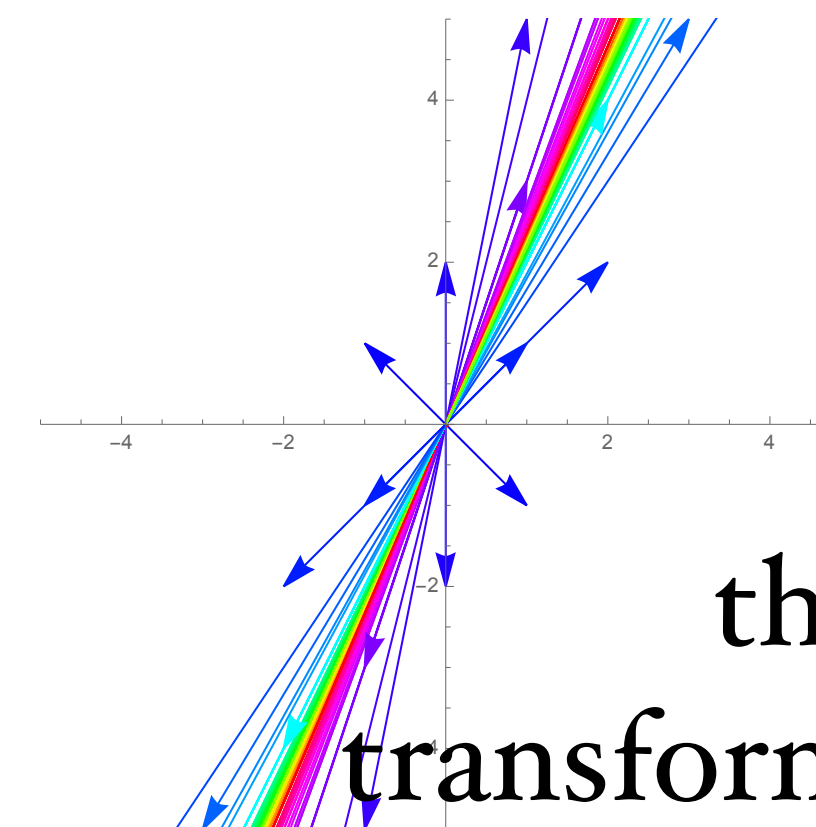
The solution is *unique*
(if it exists) if:

matrix equation

$$\begin{pmatrix} 3 & 2 \\ 7 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \text{?}$$



↓ $\begin{pmatrix} 3 & 2 \\ 7 & 4 \end{pmatrix}$



the linear
transformation is 1-to-1

pivot in every column

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

pivot in every column & row

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

pivot in every row

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linearly independent columns

basis columns

spanning columns

pivot in every column

pivot in every column & row

pivot in every row

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linearly independent columns

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$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\mathbb{R}^2 \rightarrow \mathbb{R}^3$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\mathbb{R}^3 \rightarrow \mathbb{R}^2$$

linearly independent columns

I-to-I

basis columns

I-to-I & onto

spanning columns

onto

pivot in every column

pivot in every column & row

pivot in every row

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\mathbb{R}^2 \rightarrow \mathbb{R}^3$$

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$$\mathbb{R}^3 \rightarrow \mathbb{R}^2$$

linearly independent columns

basis columns

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I-to-I

I-to-I & onto

onto

unique solution to $A\mathbf{x}=\mathbf{b}$
(if it exists, depends on $\mathbf{b}$)

unique solution to $A\mathbf{x}=\mathbf{b}$

solution to $A\mathbf{x}=\mathbf{b}$ exists

pivot in every column

pivot in every column & row

pivot in every row

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\mathbb{R}^2 \rightarrow \mathbb{R}^3$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

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$$\mathbb{R}^3 \rightarrow \mathbb{R}^2$$

linearly independent columns

basis columns

spanning columns

I-to-I

I-to-I & onto

onto

unique solution to $Ax=b$
(if it exists, depends on b)

unique solution to $Ax=b$

solution to $Ax=b$ exists

neither $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

pivot in every column

pivot in every column & row

pivot in every row

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\mathbb{R}^2 \rightarrow \mathbb{R}^3$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

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linearly independent columns

I-to-I

unique solution to $A\mathbf{x}=\mathbf{b}$
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basis columns

I-to-I & onto

unique solution to $A\mathbf{x}=\mathbf{b}$

spanning columns

onto

solution to $A\mathbf{x}=\mathbf{b}$ exists

Note: row reduction preserves all of these properties!

neither

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

Warm-up 1

a) Find two different matrices in $M_{2 \times 3}^{\mathbb{R}}$ that are in RREF and that have a pivot in each row; or explain why this is not possible.

b) Find two different matrices in $M_{3 \times 3}^{\mathbb{R}}$ that are in RREF and that have a pivot in each row; or explain why this is not possible.

square matrix with a
pivot in every column
(once put into RREF)

square matrix with a
pivot in every row
(once put into RREF)

square matrix with a
pivot in every column
(once put into RREF)



square matrix with a
pivot in every row
(once put into RREF)

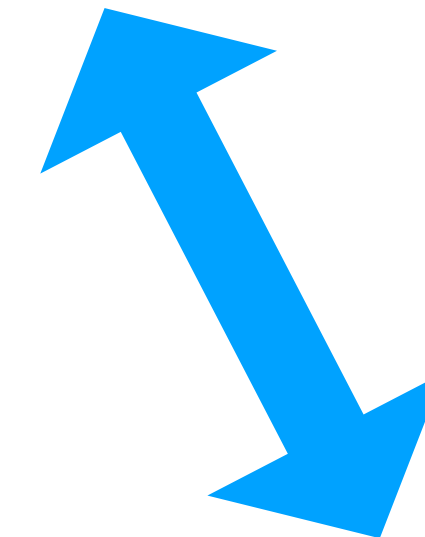
square matrix with a
pivot in every column
(once put into RREF)



square matrix with a
pivot in every row
(once put into RREF)



square matrix that represents
an 1-to-1 linear transformation



square matrix that represents
an onto linear transformation

square matrix with a
pivot in every column
(once put into RREF)

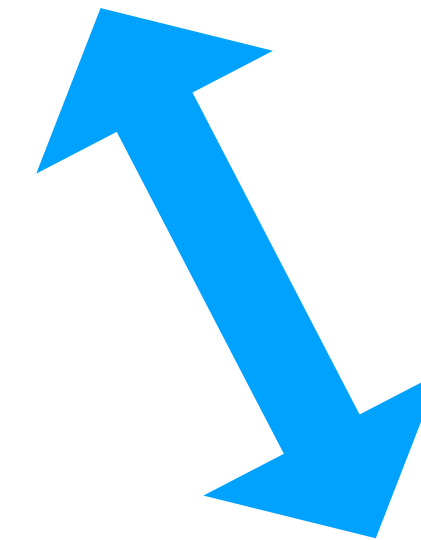
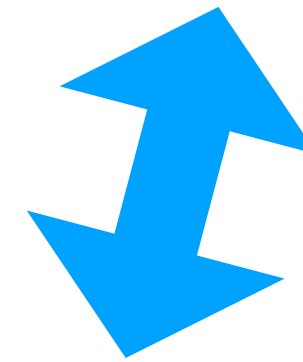
square matrix with a
pivot in every row
(once put into RREF)



square matrix that represents
an 1-to-1 linear transformation

square matrix with
spanning columns

square matrix that represents
an onto linear transformation



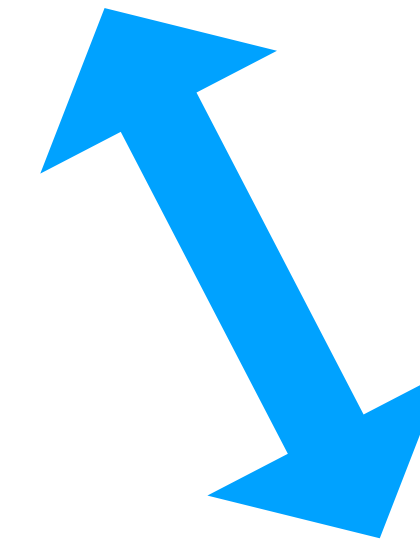
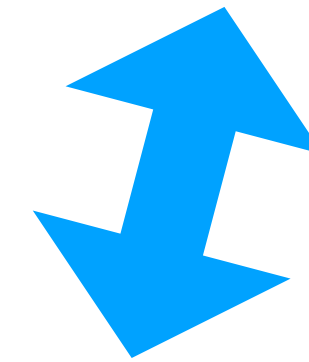
square matrix with
linearly independent
columns

square matrix with a
pivot in every column
(once put into RREF)

square matrix with a
pivot in every row
(once put into RREF)



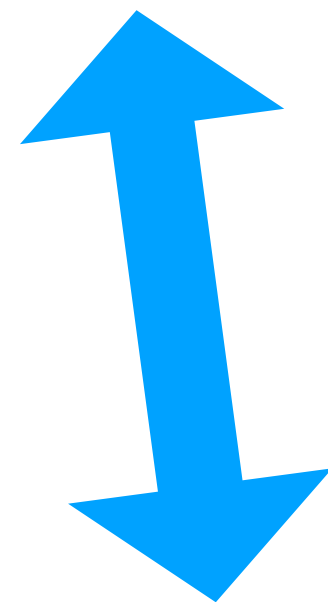
square matrix with
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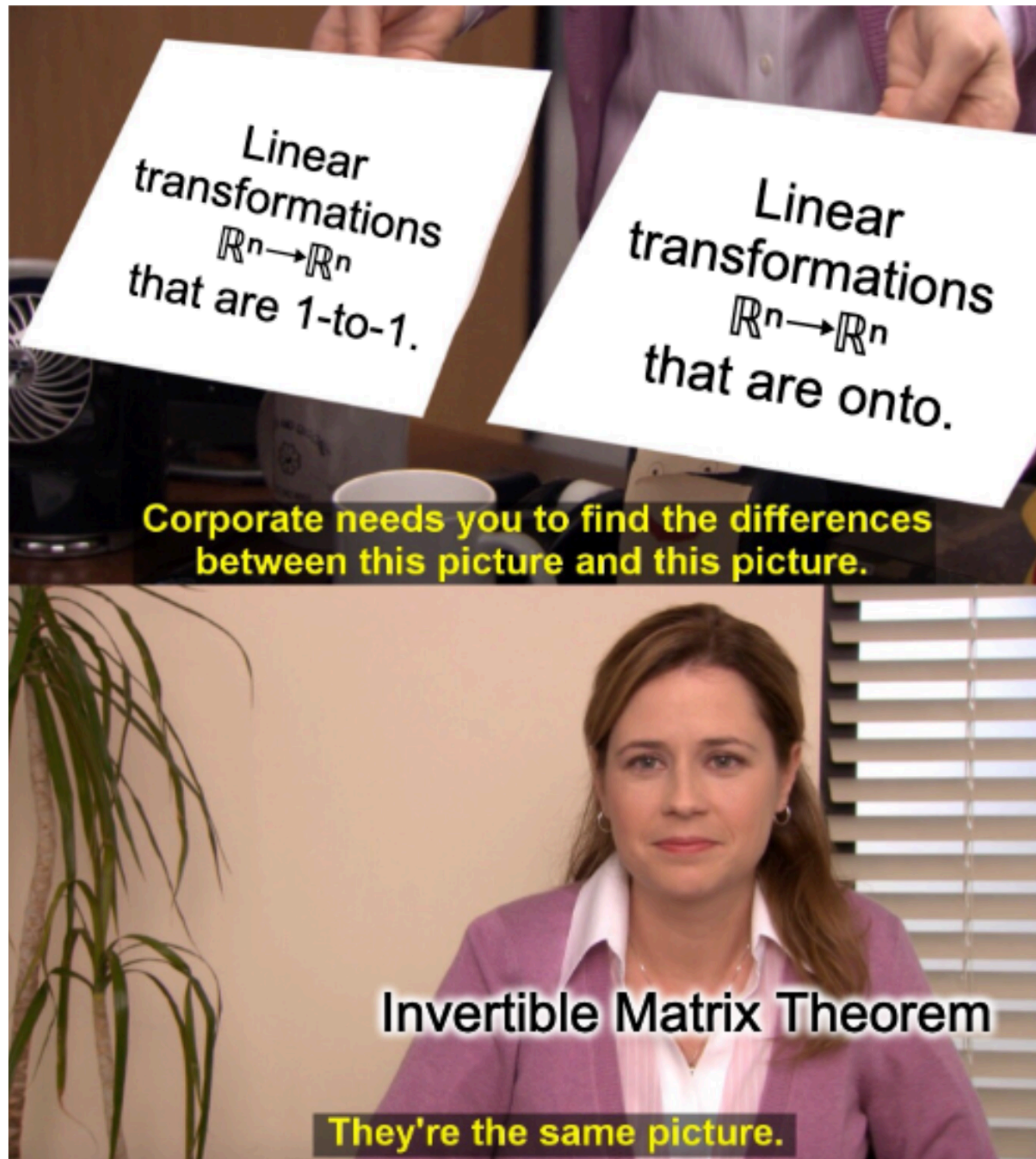
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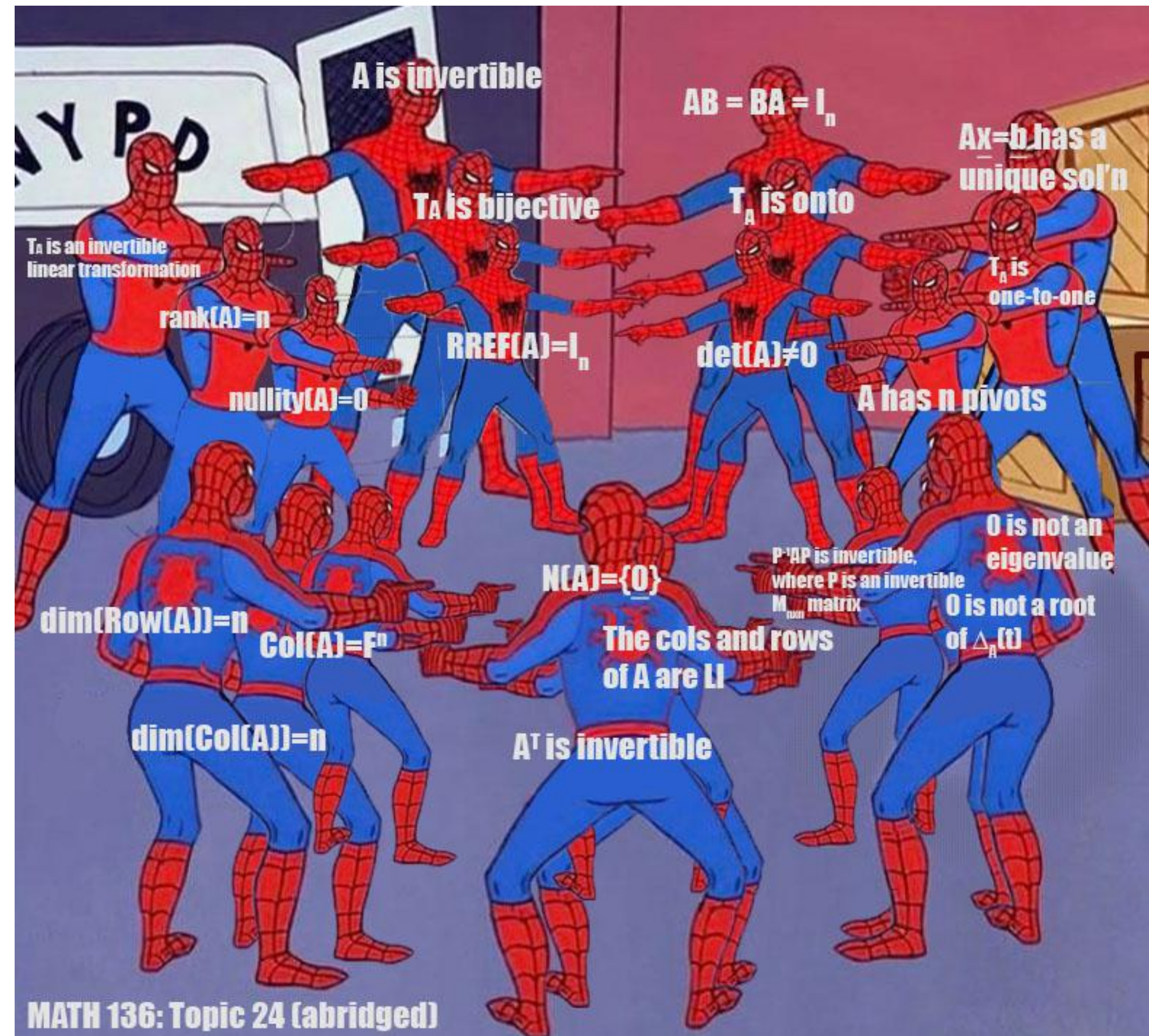
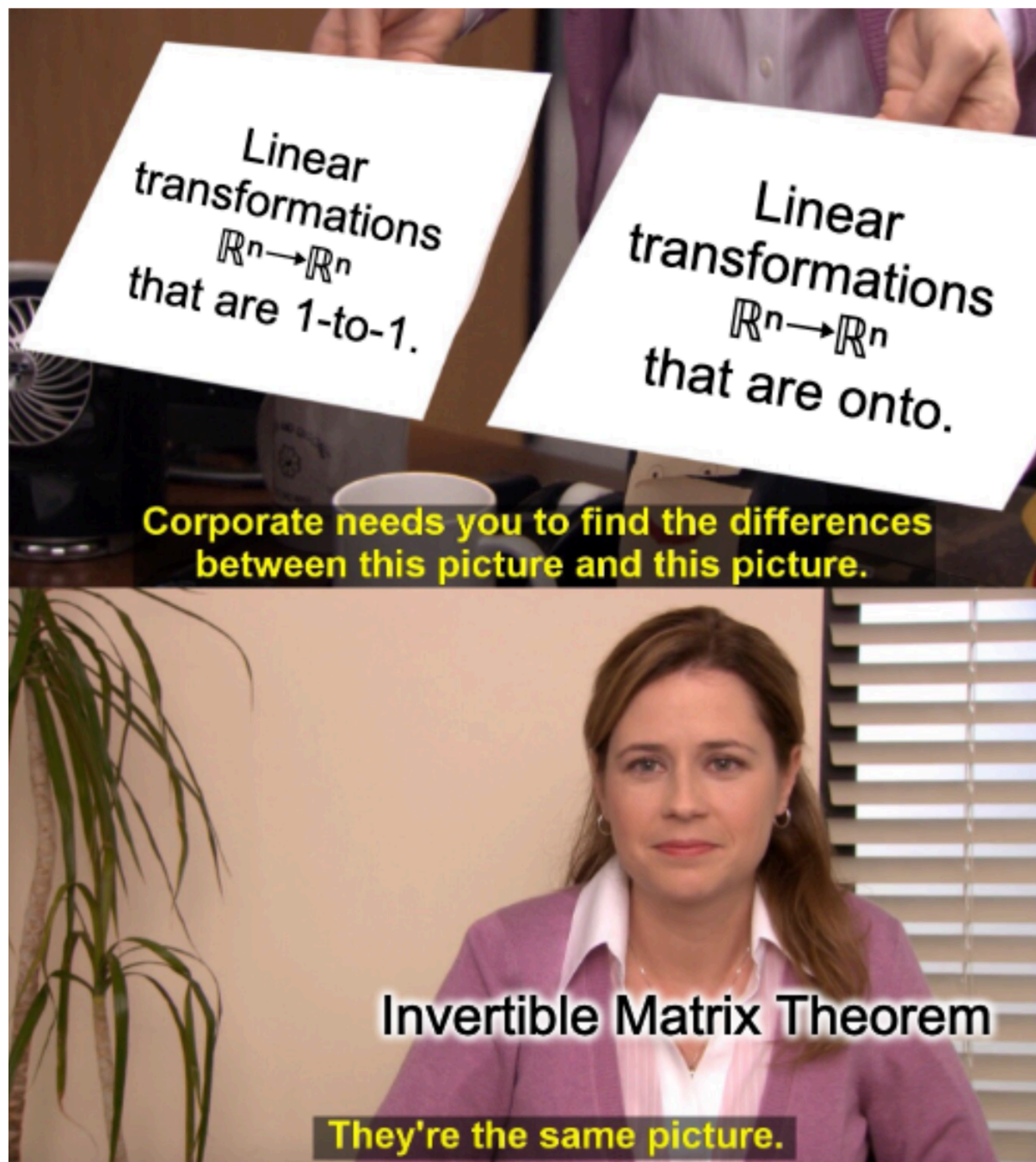


invertible square matrix

Invertible Matrix Theorem



Invertible Matrix Theorem



The Invertible Matrix Theorem. Let A be a square $n \times n$ matrix with entries in $\mathbb{F}$, and let $T : V \rightarrow V$ be the linear transformation $T(x) = Ax$. The following statements are equivalent.

- (1) A is invertible.
- (2) T is invertible.

- (3) A is row equivalent to I_n .
- (4) A has n pivots. (Prop. 2.3.6)

- (5a) $Ax = b$ has a unique solution for all b in $\mathbb{F}^n$. (Thm. 1.6.8)
- (5b) T is a bijection.
- (5c) The columns of A form a basis for $\mathbb{F}^n$. (Cor. 1.6.9)

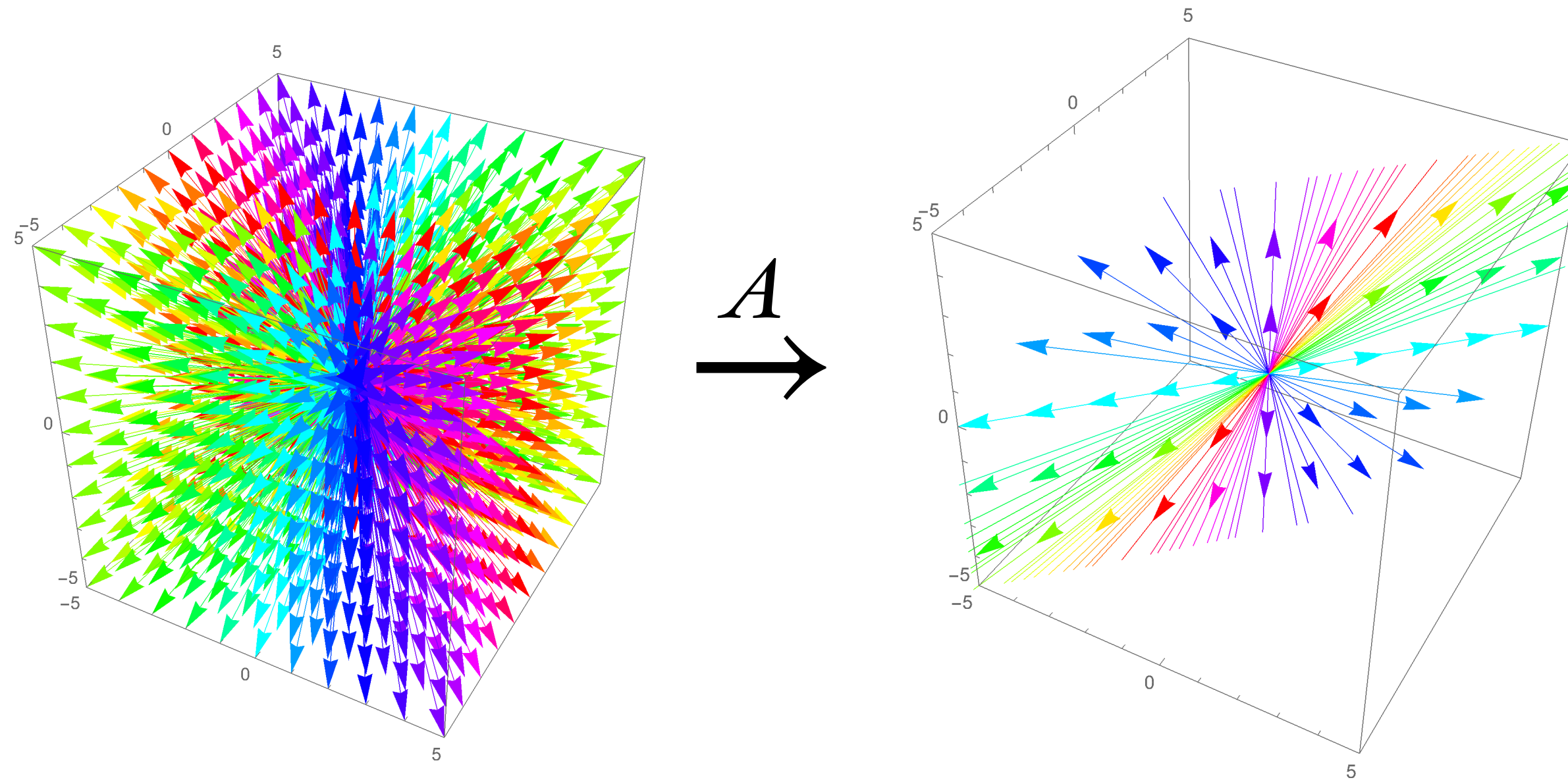
- (6a) $Ax = 0$ has only the trivial solution. ($\text{Nul } A = \{0\}$)
- (6b) T is one-to-one.
- (6c) The columns of A are linearly independent.

- (7a) $Ax = b$ is consistent for all b in $\mathbb{F}^n$.
- (7b) T is onto.
- (7c) The columns of A span $\mathbb{F}^n$. ($\text{Col } A = \mathbb{F}^n$)

- (8) $\det(A) \neq 0$.
- (9) The number 0 is not an eigenvalue of A . (*not until Chapter 4*)

Another viewpoint on row reduction

$$A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

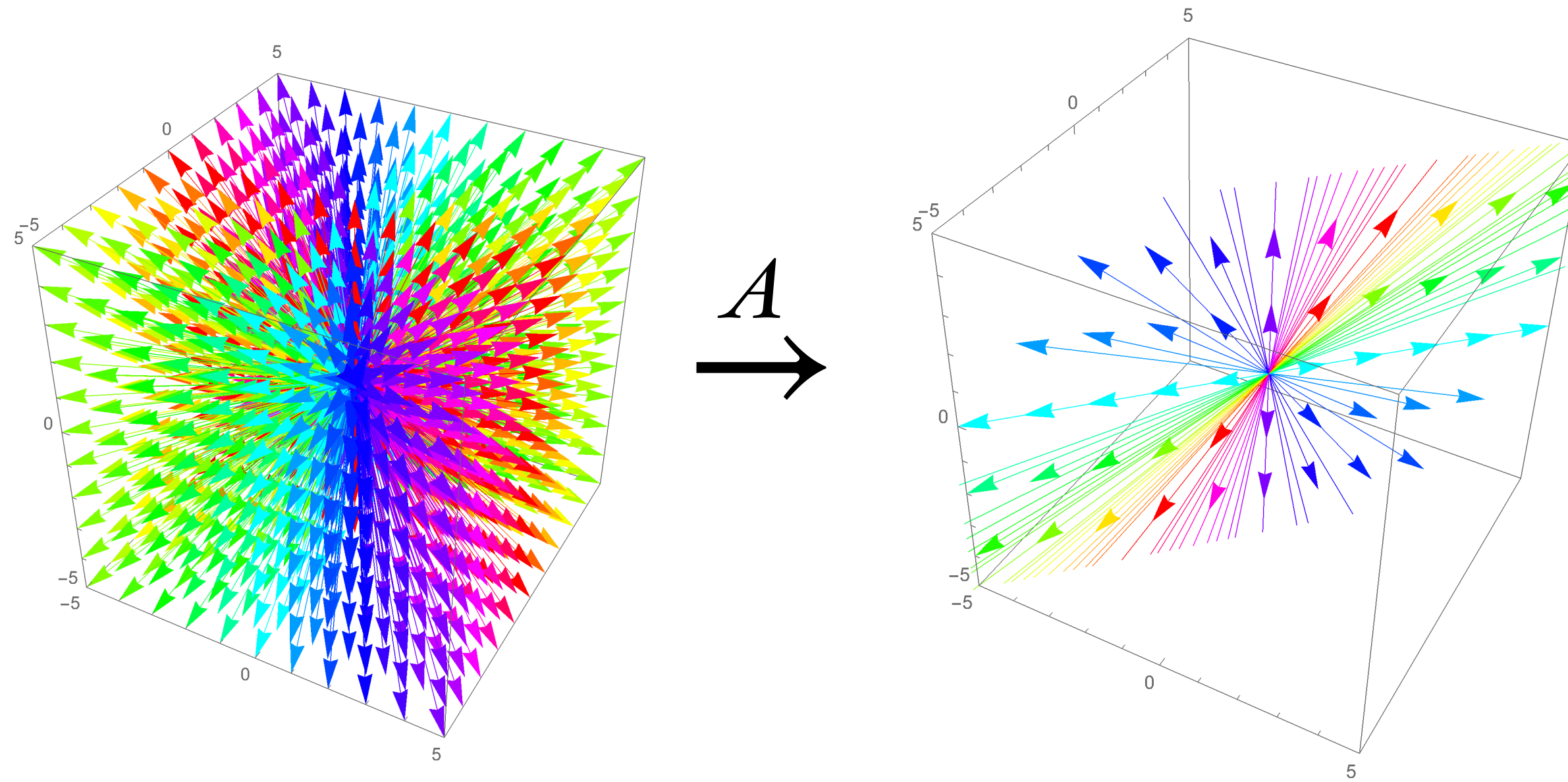


Another viewpoint on row reduction

$$A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$R : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

(row operation, invertible)

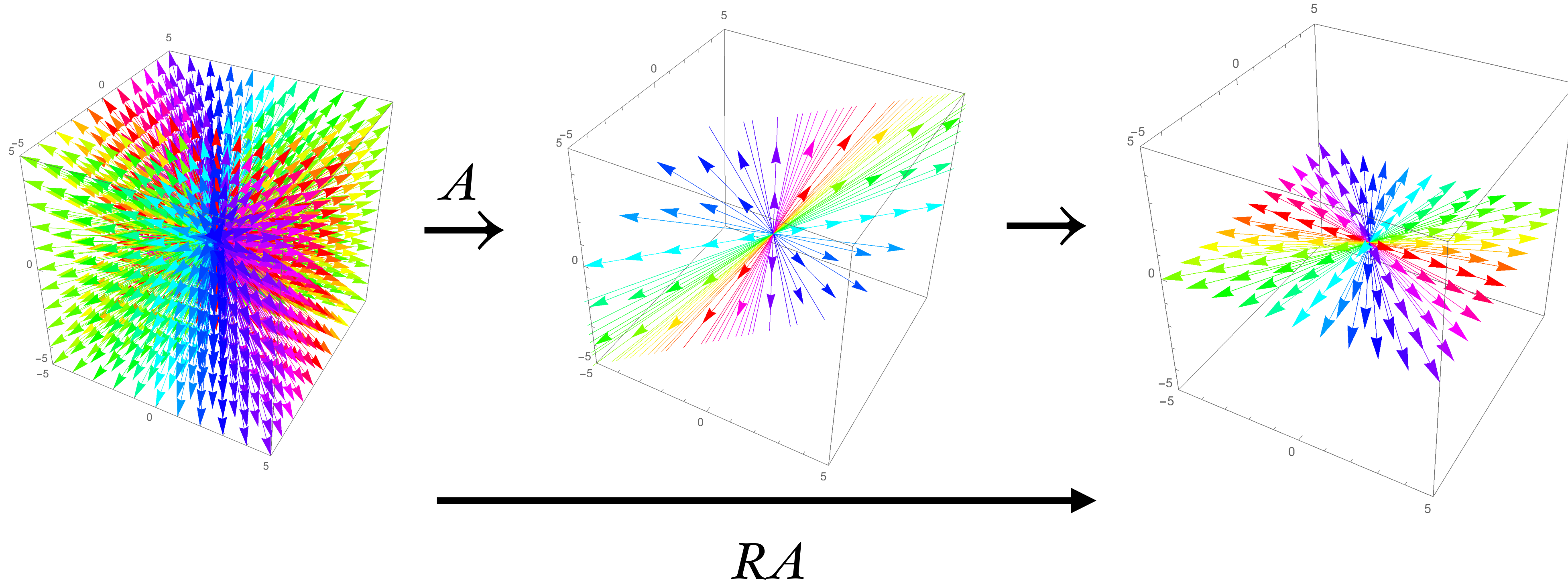


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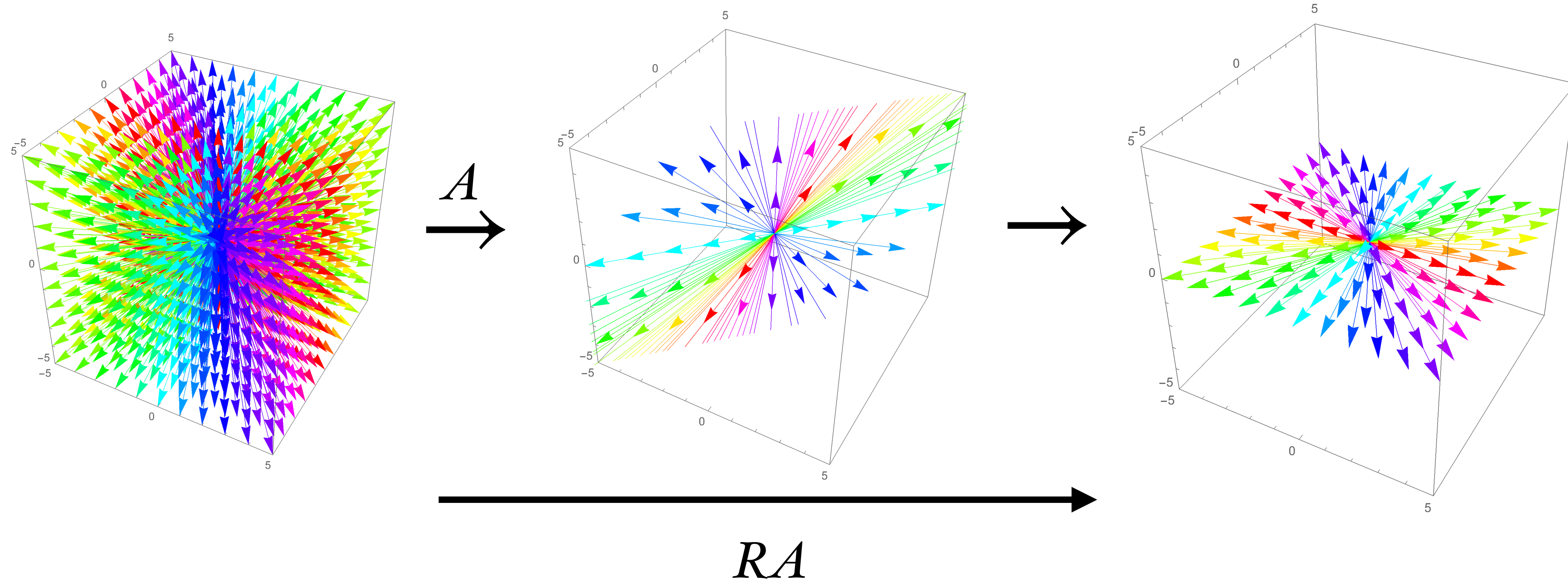


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A vs. RA : what can change? what must stay the same?

Row reduction of a matrix A ...

PRESERVES:

its nullspace (= solution set)

its nullity (= $\dim(\text{Null } A)$)

its row space

its rank (= $\dim(\text{Col } A) = \dim(\text{range})$)

whether or not columns span (or: are LI)

invertibility

CAN CHANGE:

the matrix (and corresponding LT)

its column space (= range)

the inverse

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CAN CHANGE:

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Rank-Nullity Theorem (Thm. 2.7.2): For an $m \times n$ matrix A , $\text{rank}(A) + \dim(\text{Nul}(A)) = n$.

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Pf.

number of pivot columns + number of free variable columns = number of columns ■

Row operations preserve rank and nullity.
So do column operations.

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Rank Theorem (Thm. 2.7.1). For a matrix A , $\text{rank}(A) = \text{rank}(A^T)$.

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$$\begin{pmatrix} a & e & i & m & q \\ b & f & j & n & r \\ c & g & k & o & s \\ d & h & l & p & t \end{pmatrix}$$

A

$$\begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \\ q & r & s & t \end{pmatrix}$$

A^T

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$$\begin{array}{ccccc} \left(\begin{array}{ccccc} 1 & * & 0 & * & 0 \\ 0 & 0 & 1 & * & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) & & \left(\begin{array}{cccc} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \\ q & r & s & t \end{array} \right) \\ \text{RREF of } A & & A^T \end{array}$$

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RREF of A	corresponding column operations on A^T

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$$\begin{array}{ccc}
 \begin{pmatrix} \boxed{1} & * & 0 & * & 0 \\ 0 & 0 & \boxed{1} & * & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ * & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ * & * & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{array}{c} \textcolor{blue}{RR} \\ \longleftrightarrow \end{array} & \begin{pmatrix} \boxed{1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \boxed{1} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \boxed{1} & 0 \end{pmatrix} \\
 \text{RREF of } A & \text{corresponding column operations on } A^T &
 \end{array}$$

5 things (so far!) that row reduction is great for:

- Solving a system of linear equations
- Checking a set of vectors for: linear independence, spanning, basis
- Computing the inverse of an invertible matrix

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 $A=LU$ and $A=PLU$ and $A=PLDU$ factorizations

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$$A = \begin{pmatrix} * & 0 & 0 & 0 \\ * & * & 0 & 0 \\ * & * & * & 0 \\ * & * & * & * \end{pmatrix} \begin{pmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{pmatrix}$$

lower triangular upper triangular

L U

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sometimes

always

$$A = \begin{pmatrix} * & 0 & 0 & 0 \\ * & * & 0 & 0 \\ * & * & * & 0 \\ * & * & * & * \end{pmatrix} \begin{pmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{pmatrix}$$

lower triangular

upper triangular

L

U

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \xrightarrow[\text{↔}]{R2 - \frac{1}{2}R1} \begin{pmatrix} 2 & 1 \\ 0 & \frac{1}{2} \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{array}{c} R2 - \frac{1}{2}R1 \\ \longleftrightarrow \end{array} \begin{pmatrix} 2 & 1 \\ 0 & \frac{1}{2} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ -\frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & \frac{1}{2} \end{pmatrix}$$

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$$L \quad U$$

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \xrightarrow[\text{↔}]{R2 - \frac{1}{2}R1} \begin{pmatrix} 2 & 1 \\ 0 & \frac{1}{2} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ -\frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & \frac{1}{2} \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & \frac{1}{2} \end{pmatrix}$$

$$(P)L \quad U$$

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & \frac{1}{2} \end{pmatrix} \\
 = \begin{pmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{pmatrix} \\
 \qquad \qquad \qquad \boldsymbol{L} \qquad \boldsymbol{D} \qquad \boldsymbol{U}$$

Next up:

- This week: Chapter 3 of *LADW*; Today: wrapping up Ch 2 (except 2.8).
- Quizzes 4 & 5 on Friday (finding inverses, solving systems to vector form)
- Homework 3 is due Friday / Saturday.
- HW2 debrief today 3:30-4:30pm in Eckhart 207A
- First problem swap problems due on Wednesday.